

Mahmmed Umar Faaruk

AI and Machine Learning Engineer

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Experience

Web Developer & AI Engineer

- Delivered a custom analytics dashboard for Strategic Arc Consultants using Next.js and TypeScript, replacing manual reporting workflows with a live, data-driven interface
- Rebuilt Aero Wash's website, migrating it from a legacy WordPress export to a modern, performant responsive build
- Building a cinematic, scroll-animated website for a yoga studio client, implementing custom motion design and mobile-first responsiveness
- Designed and built a QR-code-based table ordering SaaS from the ground up (database schema, backend, and customer-facing ordering flow) as both a client-ready product and portfolio piece
- Built and launched Edunox, a self-directed AI-powered study platform (React, TypeScript, Vite, Supabase, Groq), generating 15,000+ LinkedIn impressions and engagement from professionals at Oracle and Morgan Stanley; set up CI/CD via GitHub Actions

AI Developer Intern

Viswam AI (Swecha, IIIT Hyderabad and Meta)

May 2025 - Jul 2025

- Contributed to India's first Telugu Large Language Model (LLM) initiative, gaining direct exposure to NLP data pipelines, prompt engineering, and LLM fine-tuning workflows for regional language AI preservation
- Built and deployed Tune Buddy AI Chatbot by integrating an LLM inference API (Dify.ai) with a Streamlit front-end, managing end-to-end data flow from user input to personalized model response
- Architected Craft Connect on AWS (ap-south-1) and Render; designed a secure FastAPI and HTTPX API gateway ensuring no credentials or user data were exposed client-side, applying production service-hardening practices
- Implemented an offline-sync mechanism with automatic cloud synchronization to Supabase PostgreSQL, improving platform accessibility in low-connectivity environments

Web Development Intern

Intrainz

Oct 2024-Dec 2024

- Designed and delivered three functional JavaScript web applications including a calculator, e-commerce store-front, and task manager, applying DOM manipulation, state management, and mobile-first responsive design principles
- Built a personal portfolio site from scratch using semantic HTML, CSS, and JavaScript, demonstrating end-to-end project delivery and technical documentation skills

Projects

Fire and Smoke Detection System (*Published Research*)

Tools: PyTorch, YOLOv11, Python, OpenCV, Twilio SMS API

- Designed and trained a real-time computer vision pipeline for fire and smoke detection achieving 93.5% mAP accuracy at 60 FPS, suitable for production safety-monitoring environments
- Built a complete data pipeline covering ingestion, preprocessing, augmentation, labelling, and model evaluation; benchmarked YOLOv11 against multiple baseline algorithms using mAP, F1-Score, Precision, and Recall metrics
- Integrated automated SMS incident alerting via Twilio API to simulate a real-time model monitoring and incident-response workflow
- Research accepted for publication in an AI Healthcare book (2025, In Press)

Adaptive AI-SIEM Federated Cyber Threat Detection

Tools: PyTorch, CNN, LSTM, Federated Learning, Django, TF-IDF, Blockchain

- Built a multi-model deep learning system (CNN and LSTM) trained on benchmark datasets NSL-KDD and CICDS2017, achieving 99% accuracy and 0.01 False Positive Rate
- Integrated Federated Learning and Blockchain for distributed model training across IoT edge nodes without sharing raw data

Craft Connect Cloud-Integrated Full-Stack Platform

Tools: Python, FastAPI, Supabase, PostgreSQL, AWS, Streamlit, JavaScript

- Architected and deployed a mobile-first platform on AWS with a FastAPI backend and PostgreSQL via Supabase; integrated Google APIs with an external corpus API for verified contributions
- Engineered a custom offline-sync mechanism with automatic cloud synchronization, optimized for low-connectivity environments

Chess Buddy AI-Powered Web Application

Tools: Python, Streamlit, Stockfish Engine

- Integrated the Stockfish chess engine into a Python backend for real-time move computation; designed an interactive Streamlit UI managing board states, rule enforcement, and outcome detection

Technical Skills

Languages: Python, SQL (basic)

ML & Data Libraries: scikit-learn, NumPy, Pandas, Matplotlib, PyTorch, CNN, LSTM, Federated Learning, Transfer Learning Supervised & Unsupervised Learning, Classification, Regression, NLP (TF-IDF, Tokenization, Word Embeddings), Computer Vision, Object Detection (YOLOv11, OpenCV)

Generative AI & LLMs: Prompt Engineering, Lang Chain, Dify.ai, LLM Fine-Tuning Workflows

Frontend & Full-Stack: React, Next.js, TypeScript, Vite, JavaScript, HTML, CSS, Tailwind CSS, Firebase **Cloud & Deployment:** AWS (ap-south-1), Render, Streamlit Cloud, FastAPI, Docker (familiar)

Databases: PostgreSQL, Supabase, Google Firestore

Dev Practices: Git, REST API design, unit testing fundamentals, technical documentation

Education

B.Tech., Computer Science and Engineering (AI and ML) July 2025
Sreyas Institute of Engineering and Technology, Hyderabad | CGPA: 6.28

Certifications and Publications

Certifications:

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|---|-----------------|
| • AWS APAC Solutions Architecture Virtual Experience - Forage | <i>Jan 2025</i> |
| • Generative AI Workshop -Growth School | <i>Mar 2026</i> |
| • Machine Learning Workshop - Innomatics Research Labs | <i>Nov 2025</i> |
| • Goldman Sachs Software Engineering Virtual Experience -Forage | <i>Jul 2025</i> |
| • Tata Data Visualization Job Simulation - Forage | <i>Jul 2025</i> |
| • Machine Learning Workshop - IEEE Sreyas | <i>Dec 2023</i> |

Publication:

- Employing Deep Learning Paradigms for Fire and Smoke Detection (Accepted, AI Health Care book, 2025) (In Press)

Technical Articles:

- NumPy Basics: A Practical Guide to Understanding NumPy with Real-World Case Studies (2026)
- Mastering Python Functions (2026)